

# Post-Course Assessment

Complete after the capstone. Use evidence from your work when explaining answers.

Student: \_\_\_\_\_

Date: \_\_\_\_\_

Pathway: Code Learning / Robot Programming

Team: \_\_\_\_\_

Scoring: Questions 1-6 are worth 1 point each. Questions 7-12 are worth 2 points each. Total: 18 points.

1. Choose the best type for each value: motor port, measured distance with decimals, and whether a piston is extended.

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2. Explain the error in `if (finished = true)` and repair it.

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3. Describe what happens while the condition in a `while` loop remains true.

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4. Explain how a declaration in a header connects to a definition in a source file.

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5. Name the source file that creates the template's VEX motors and sensors.

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6. In `move(75, 180, 7)`, explain what 180 controls.

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7. Describe one movement error caused by an incorrect software gear ratio.

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8. If target error doubles, what happens to the proportional contribution when  $K_p$  stays constant? Explain.

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9. Describe one condition that can cause integral windup and how the template limits it.

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10. Why is a ten-run reliability test stronger evidence than one successful run?

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11. Name two physical or environmental variables that should be recorded during autonomous testing.

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12. Give one project need that would justify changing to an odometry-based framework.

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## Results

Final score: \_\_\_\_\_ / 18 Initial score: \_\_\_\_\_ / 18 Percentage-point improvement: \_\_\_\_\_

Confidence after the course (1-5): \_\_\_\_\_ Capstone score: \_\_\_\_\_ / 100 Trial success rate: \_\_\_\_\_ %